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- a.
 1. The training error is 28.65
 2. Predicted test error is 32.34
- b. I started with a simple degree-1 linear regression model using the original eight features. Since both the training MSE and cross-validated MSE were relatively high, I explored polynomial feature expansions. I added second-, third-, and fourth-degree terms and tested higher polynomial degrees for each variable individually. I compared both training MSE and cross-validated MSE to determine whether increasing the degree of a particular variable improved the model.
I then expanded the feature set further by introducing nonlinear transformations of the original variables, including quadratic, logarithmic, square-root, sine, cosine, and reciprocal functions such as $1/(1+|x|)$ and $1/(1+|x|)$. This produced an initial nonlinear basis containing 57 features. The expanded model substantially improved performance, reducing the cross-validated MSE to approximately 36.6.
Since the model now contained many features, I next performed feature reduction. I tested the effect of removing features and compared both the resulting training MSE and cross-validated MSE. I then used sequential backward elimination, removing one feature at a time when its removal improved cross-validated performance. This reduced the feature set while improving the cross-validated MSE from approximately 36.67 to 34.98. Finally, I considered pairwise interaction terms of the form $x_i x_j$, which allow the effect of one variable to depend on another. I tested the candidate interactions by examining their effect on both training MSE and cross-validated MSE, and retained only interactions that improved cross-validated performance. Although this increased the number of features again, it improved both errors and produced the final 49-feature model, with a training MSE of approximately 28.65 and a 5-fold cross-validated MSE of approximately 32.35.
- c. We use the 5 fold cross validation method to calculate the predicted test error. The process is essentially to divide the training data into 5 folds. Train on 4 of them and test on the 5th fold and repeat this for every fold and take the average of the test folds error.
- d. We used cross validation and feature selection for this. We started with a larger non linear feature set and evaluated using both cv mse and mse. We then removed features that didn't improve our cv mse and we also tested higher order terms and removed them as they were increasing our cv mse. Finally we added the interaction terms only when they decreased the cv mse.